PROJECT S.M.A.L.L.

SWARM MICRO-FARMING AUTONOMOUS LOCALIZED LOGISTICS

Plant-by-plant precision agriculture, delivered by a low-cost autonomous robot swarm — fully off-grid.

Edge AI | ESP-NOW Mesh | 5-Robot Swarm | Zero Cloud Dependency

Big precision. Tiny footprint.

Autonomous · Scalable · Off-Grid

THE PROBLEM

Macro-farming treats an entire field as one uniform unit

💧 Water Waste

Blanket irrigation floods entire plots regardless of individual plant need.

☣️ Chemical Runoff

Uniform pesticide & fertilizer spraying pollutes soil and waterways.

💰 High Equipment Cost

Heavy industrial machinery is expensive to buy, run, and maintain.

📊 Inefficient Management

No plant-level data means yield and health issues go undetected.

THE VISION: MICRO-FARMING

Continuous, plant-specific analysis and milliliter-exact treatment — delivered by a coordinated swarm of low-cost autonomous robots.

MACRO → MICRO

✗ Whole-field blanket spraying	✓ Root-zone, plant-specific dosing
✗ Fixed schedule irrigation	✓ Real-time soil-moisture triggers
✗ Manual scouting	✓ Continuous CV-based monitoring
✗ Heavy tractors / large capex	✓ Swarm of low-cost micro-robots

Key Numbers

5	Robots per swarm
mL	Level dosing precision
24/7	Autonomous operation
0	Cloud / cellular dependency

SYSTEM ARCHITECTURE

A distributed master / node design solves the compute-vs-power constraint in remote fields

Raspberry Pi Edge Server

Central Brain — Vision & Decisions

Off-grid processing hub running edge AI models for plant-health, weed, and moisture detection locally.

ESP32-CAMs

Sensing & Capture Nodes

Mounted per robot; capture high-res crop images and stream raw visuals to the Pi — no local CV compute.

ESP32 Controllers

Actuation & Sensor Layer

Real-time motor control, capacitive soil-moisture readings, and ultrasonic obstacle avoidance.

Battery-Swap Hub

24/7 Autonomy

Central dock where robots auto-swap depleted cells for charged ones — continuous unassisted operation.

THE CENTRAL BRAIN

Raspberry Pi Edge Server

Off-grid processing hub

Runs entirely on-site — no cloud services, no internet dependency.

Computer-vision models on the edge

Hosts plant-health / weed / moisture detection models locally.

Central decision-making

Converts image + sensor data into precise dosing instructions per plant.

Mesh network endpoint

Aggregates image + telemetry data relayed from the entire robot swarm.

Edge Compute Specs

Quad-core ARM CPU

Full processing power for on-device inference and coordination

On-device inference (TFLite / ONNX)

Run plant detection and decision models without cloud

GPIO ⟶ ESP-NOW gateway

Bridge between Pi and the robot swarm mesh network

Local storage for field logs

All data stays on-site — privacy, reliability, offline access

SENSING & ACTUATION NODES

Each robot pairs two ESP32-class microcontrollers with distinct jobs

ESP32-CAM

Sensing & Capture Node

- Captures high-resolution crop imagery
- Streams raw visual data to the Pi over the mesh
- No onboard CV — offloads heavy inference to the central brain

ESP32 Controller

Actuation & Sensor Controller

- Real-time motor / drivetrain control
- Capacitive soil-moisture sensing
- Ultrasonic depth sensing for obstacle avoidance

24/7 AUTONOMY — THE BATTERY-SWAPPING HUB

Central dock enabling continuous, unassisted field operation with no human intervention

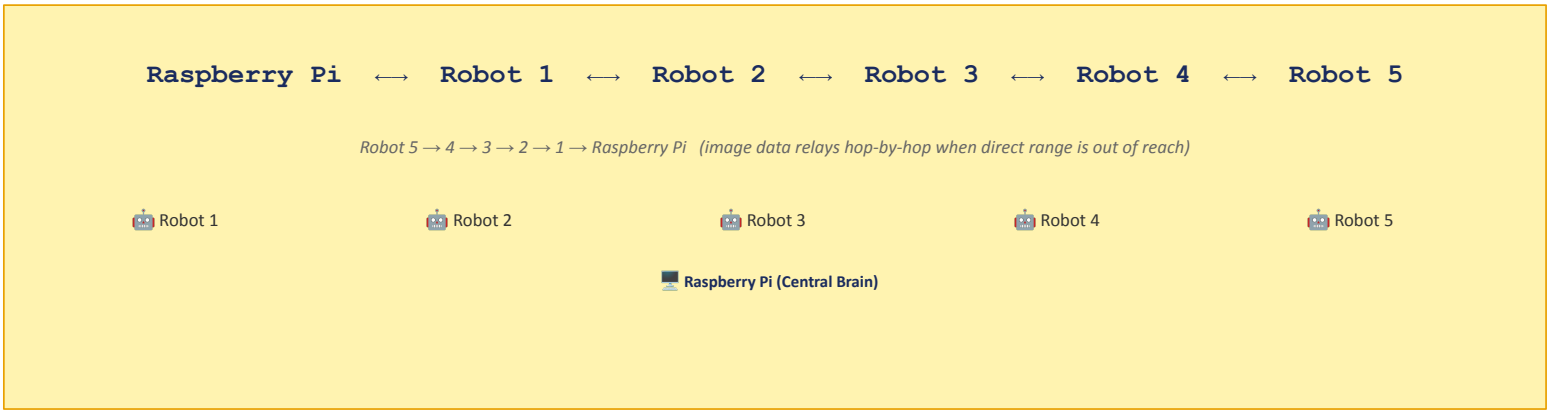
1	2	3	4
Robot returns low-battery	Auto-swap at docking hub	Depleted cell recharges	Robot redeploys full power

5	< 2 min	24/7	0
Docking bays for the full swarm	Estimated swap time	Continuous field coverage	Human interventions needed

Result: robots never sit idle waiting on a charge cable — the swarm maintains near-continuous field coverage around the clock.

SWARM INTELLIGENCE & NETWORKING

Peer-to-peer multi-hop ESP-NOW mesh — no cellular, no Wi-Fi router required



Decentralized Relay

Out-of-range robots relay image data through adjacent nodes instead of requiring direct server contact.

Self-Healing Topology

If a node is blocked or drops offline, the mesh dynamically reroutes packets through nearby active nodes.

DEEPTECH NOVELTY & VALUE

Why S.M.A.L.L. is more than another farm robot

Hardware-Software Fusion

Pairs lightweight edge nodes with centralized edge AI over a low-power mesh — solving real embedded R&D bottlenecks.

Complete Off-Grid Autonomy

Zero reliance on cloud APIs, 4G/5G coverage, external infrastructure, or manual human intervention.

Democratic Precision Farming

Unlocks plant-level analytics and localized hydration at a fraction of industrial machinery cost.

Precision agriculture, once reserved for large industrial farms, becomes accessible to smallholders everywhere.

WHAT'S NEXT

From prototype to field-ready swarm

Phase 1

Hardware Bring-Up

Single-robot prototype: ESP32-CAM + controller + Pi pipeline validated end-to-end.

Phase 2

Mesh at Scale

Extend to full 5-robot ESP-NOW mesh; validate self-healing multi-hop relay in field conditions.

Phase 3

Autonomous Ops

Deploy battery-swap hub for unattended 24/7 operation across a full growing season.

Phase 4

Field Pilot

Real farm pilot measuring water/chemical savings vs. traditional macro-farming baseline.

Thank you · PROJECT S.M.A.L.L. · Big precision. Tiny footprint.